

Chapter B3: Living together – food and ecosystems

Overview

All living organisms depend on the ability of photosynthetic organisms to synthesise glucose from carbon dioxide and water in the presence of light, and on feeding relationships to transfer biomass through communities.

From study at earlier Key Stages, learners will be familiar with the reactants and products of photosynthesis, and the need for light in the process. In Topics B3.1 and B3.2 the context of photosynthesis is used to explore several

fundamental concepts in biology, including enzyme action and the movement of substances by diffusion, osmosis and active transport.

Learners expand their knowledge of the interdependencies between organisms within ecosystems in Topic B3.3, through understanding of food webs, competition for resources, and the cycling of substances.

Finally, Topic B3.4 considers the effects that environmental changes and human activities can have on interacting populations within ecosystems.

Learning about food and ecosystems before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- understand the similarities and differences between plant and animal cells
- know that some organisms make their own food using photosynthesis
- know that photosynthesis in plant cells occurs in the chloroplasts
- know the reactants in, and products of, photosynthesis, and be able to write a word summary
- know that photosynthesis requires light
- be familiar with the adaptations of leaves for photosynthesis, and the role of stomata in gas exchange
- know that water and minerals enter a plant through the roots
- know that molecules of a solute move through solvent, and through cell membranes, by diffusion
- know that animals obtain their food from plants (and other animals that ate plants)
- understand the difference between carnivores, herbivores and omnivores, and between producers and consumers
- know that individuals of the same type living in the same place make up a population, and that all the interacting populations in an ecosystem make up the community
- understand the use of food chains and food webs as models of the feeding relationships within a community
- appreciate the interdependence of organisms in a community, including food webs, the breakdown and cycling of substances, and animals as pollinators
- know that changes in an ecosystem can affect the survival of individuals and populations.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements

will be assessed in both Foundation and Higher Tier papers.

Learning about food and ecosystems at GCSE (9–1)

B3.1 What happens during photosynthesis?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Producers make glucose using photosynthesis. Some of the glucose is used as the fuel for cellular respiration, some is converted into starch and then stored, and the rest is used to make lipids, proteins and other carbohydrates for growth. Photosynthesis involves many chemical reactions, but can be summarised in two main stages. The first stage requires light and chlorophyll (located in chloroplasts in plant cells) to split water molecules into hydrogen and oxygen. The hydrogen is transferred to the second stage, but the oxygen is released into the atmosphere as a waste product. The second stage combines carbon dioxide with hydrogen to make glucose.	- describe the process of photosynthesis, including the inputs and outputs of the two main stages and the requirement of light in the first stage, and describe photosynthesis as an endothermic process - describe practical investigations into the requirements and products of photosynthesis <i>PAG5</i> - explain how chloroplasts in plant cells are related to photosynthesis
The reactions in photosynthesis and many other biological processes are catalysed by enzymes. The lock and key model can be used to explain enzyme action. It can also be used to make predictions about the effect on the rate of enzyme-catalysed reactions when the substrate concentration, temperature and pH are changed (1aS3).	- explain the mechanism of enzyme action including the active site, enzyme specificity and factors affecting the rate of enzyme-catalysed reactions, including substrate concentration, temperature and pH - describe practical investigations into the effect of substrate concentration, temperature and pH on the rate of enzyme controlled reactions <i>M2b, M2f, M4a, M4b, M4c PAG4</i>

Linked learning opportunities

Practical work:

- on a whole plant, wrap one leaf in foil, and enclose another leaf in a conical flask with a small amount of KOH (to remove CO₂); after 24h, test leaves for starch

Practical work:

- investigate effects of substrate concentration, temperature and pH on enzyme activity

Ideas about Science:

- lock and key model to explain and make predictions about enzyme activity (1aS3)

B3.1 What happens during photosynthesis?

Teaching and learning narrative

Understanding of how factors affect enzyme activity helps to explain the effects of temperature and carbon dioxide concentration on the rate of photosynthesis. The effect of light intensity is explained by the need for light to bring about reactions in photosynthesis. **Light intensity is inversely proportional to the square of the distance from the light source (the inverse square law); this helps us explain why the rate of photosynthesis changes in the way that it does with distance from a point light source.**

Assessable learning outcomes

Learners will be required to:

4.
 - a) explain the effect of temperature, light intensity and carbon dioxide concentration on the rate of photosynthesis
 - b) describe practical investigations into the effect of environmental factors on the rate of photosynthesis
PAG5
5. **use the inverse square law to explain changes in the rate of photosynthesis with distance from a light source**
6. **explain the interaction of temperature, light intensity and carbon dioxide concentration in limiting the rate of photosynthesis, and use graphs depicting the effects**
7. in the context of the rate of photosynthesis:
 - a) understand and use simple compound measures such as the rate of a reaction
M1a, M1c
 - b) translate information between graphical and numerical form
M4a
 - c) plot and draw appropriate graphs selecting appropriate scales for axes
M4a, M4c
 - d) extract and interpret information from graphs, charts and tables
M2c

Linked learning opportunities

Practical work:

- investigate rate of photosynthesis by collecting gas or counting bubbles from pondweed
- use a datalogger to measure oxygen concentration, pH, temperature and light intensity over 24h for pondweed

B3.2 How do producers get the substances they need?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
The ways in which photosynthetic organisms take in carbon dioxide and water for photosynthesis, and release the waste product oxygen, illustrate the principles of diffusion and osmosis. Generally, molecules move from a region of their higher concentration to a region of their lower concentration; the difference in concentration drives a change towards equal concentration. Carbon dioxide and oxygen molecules move by diffusion, through cell membranes in single-cellular (prokaryotic) producers, and through stomata and cell membranes in plants. Water molecules move by osmosis through cell membranes; projections from root cells ("root hairs") of plants increase the surface area for osmosis. The way in which photosynthetic organisms take in nitrogen (to make proteins) illustrates the process of active transport. Producers get nitrogen from nitrate ions (NO_3^-). Molecules of water and gases can diffuse through partially-permeable cell membranes but nitrate ions cannot; producers use energy from molecules of ATP to transport nitrate ions through the cell membrane by active transport.	1. describe some of the substances transported into and out of photosynthetic organisms in terms of the requirements of those organisms, including oxygen, carbon dioxide, water and mineral ions
	2. a) explain how substances are transported into and out of cells through diffusion, osmosis and active transport b) describe practical investigations into the processes of diffusion and osmosis <i>PAG8</i> <i>① Learners are not expected to explain osmosis in terms of water potential</i>
	3. explain how the partially-permeable cell membranes of plant cells and prokaryotic cells are related to diffusion, osmosis and active transport
	4. explain how water and mineral ions are taken up by plants, relating the structure of the root hair cells to their function

Linked learning opportunities

Practical work:

- investigate diffusion using drops of ink in water and in agar in Petri dishes on graph paper
- investigate diffusion across a partially permeable membrane using starch suspension in dialysis tubing in a beaker of water; compare adding iodine solution inside versus outside the tubing
- investigate the effect of solute concentration on osmosis using potato cylinders in sugar solution

B3.2 How do producers get the substances they need?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Plants do not have blood to transport substances around the organism; they have transport vessels formed from xylem and phloem. Water and ions (e.g. nitrate) in aqueous solution are moved through xylem from the roots and up the stem/trunk by transpiration, to replace water that evaporates from the plant surface and diffuses out of open stomata. Sugars are moved through phloem from photosynthetic to non-photosynthetic tissues by translocation. Sugars are loaded into phloem by active transport, then water moves into the concentrated solution by osmosis and pushes the substances along the tube. The rate of water uptake by a plant can be affected by environmental factors. Light intensity and temperature affect the rate of photosynthesis (and therefore the demand for water), while air movement and temperature affect the rate of water loss from aerial parts of the plant.	5. a) explain how the structure of the xylem and phloem are adapted to their functions in the plant b) describe how to use a light microscope to observe the structure of the xylem and phloem <i>PAG1</i>
	6. a) describe the processes of transpiration and translocation, including the structure and function of the stomata b) describe how to use a light microscope to observe the structure of stomata <i>PAG1</i> c) describe how to use a simple potometer <i>PAG6</i> ① <i>Learners are not expected to describe transpiration in terms of tension or pressure, and are not expected to describe translocation in terms of water potential or hydrostatic pressure</i>
	7. a) explain the effect of a variety of environmental factors on the rate of water uptake by a plant, to include light intensity, air movement, and temperature b) describe practical investigations into the effect of environmental factors on the rate of water uptake by a plant <i>PAG6</i>
	8. in the context of water uptake by plants: a) use simple compound measures such as rate M1a, M1c b) carry out rate calculations M1a, M1c c) plot, draw and interpret appropriate graphs M4a, M4b, M4c, M4d d) calculate percentage gain and loss of mass M1c

Linked learning opportunities

Practical work:

- use eosin stain to observe xylem in broad bean plant stem under hand lens and microscope
- observe stomata (paint two thin layers of nail varnish onto a leaf, put clear tape over then peel off, stick to microscope slide)

B3.3 How are organisms in an ecosystem interdependent?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Producers take in carbon and nitrogen compounds from their environment and use them (along with oxygen, hydrogen and other elements) to make small organic molecules including sugars, fatty acids, glycerol and amino acids. These small molecules are used to make larger organic molecules, such as long-chain carbohydrates, lipids and proteins. The larger molecules are used to build new structures (e.g. membranes, organelles). Consumers can only get their supply of carbon and nitrogen compounds by eating producers (or other consumers that ate producers) and digesting the biomass. This releases the small molecules so they can be absorbed and then used to build biomass in the consumer. The transfer of biomass between organisms is one way in which the populations in a community are interdependent, and can be modelled using a food web (1aS3). The amount of biomass present at each trophic level is not shown by a food web, but can be modelled using a pyramid of biomass (1aS3). The size of each population in a community is limited by predation and competition for food and other resources including space, water, light, shelter, mates, pollinators and seed dispersers.	- explain the importance of sugars, fatty acids and glycerol, and amino acids in the synthesis and breakdown of carbohydrates, lipids and proteins - describe the use of qualitative tests for biological molecules <i>PAG2</i> - describe photosynthetic organisms as the main producers of food and therefore biomass for life on Earth - describe some of the substances transported into organisms in terms of the requirements of those organisms, including dissolved food molecules - describe different levels of organisation in an ecosystem from individual organisms to the whole ecosystem - explain the importance of interdependence and competition in a community - describe the differences between the trophic levels of organisms within an ecosystem <i>(separate science only)</i> - describe pyramids of biomass and explain, with examples, how biomass is lost between the different trophic levels <i>(separate science only)</i> - calculate the efficiency of biomass transfers between trophic levels and explain how this affects the number of organisms at each trophic level <i>(separate science only)</i> M1c

Linked learning opportunities

Practical work:

- investigate the breakdown of starch into sugars using amylase and test strips

Ideas about Science:

- use a food web as a model to explain interdependence in a community, identify limitations of the model, and use it to make predictions about the effects that a change in the ecosystem could have on the interacting populations (1aS3)
- pyramids of biomass as models of biomass transfer in a food chain (1aS3)

B3.3 How are organisms in an ecosystem interdependent?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Substances essential to life, including water and carbon, cycle through the biotic and abiotic components of ecosystems so that they can be used and reused by organisms. Water cycles through precipitation, food chains, transpiration, excretion, runoff, flow through streams/ rivers/oceans, and evaporation. Carbon cycles through photosynthesis, food chains, cellular respiration, decomposition and combustion. Decomposition is catalysed by enzymes released by microorganisms. Rate of decomposition is affected by environmental factors: temperature affects enzymes and the rate of reactions; microorganisms need water to survive and many need oxygen for aerobic respiration. Landfill sites are often oxygen deficient, leading to an increase in anaerobic decomposition which produces methane – a gas with a much greater greenhouse effect than the carbon dioxide produced by aerobic decomposition.	9. recall that many different substances cycle through the abiotic and biotic components of an ecosystem, including carbon and water
	10. explain the importance of the carbon cycle and the water cycle to living organisms
	11. explain the role of microorganisms in the cycling of substances through an ecosystem
	12. calculate the percentage of mass, in the context of the use and cycling of substances in ecosystems M1c
	13. explain the effect of factors such as temperature and water content on rate of decomposition in aerobic and anaerobic environments (<i>separate science only</i>)
	14. calculate rate changes in the decay of biological material (<i>separate science only</i>) M1c

Linked learning opportunities

Practical work:

- culture microorganisms on starch agar, stain with iodine solution; clear areas beyond cultures show digestion by extracellular amylase

B3.4 How are populations affected by conditions in an ecosystem?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
The distribution and abundance of organisms in an ecosystem depends on abiotic and biotic factors. The size of one or more populations in a community may be affected if the environmental conditions change, or if a new substance, competitor, predator or pathogen is introduced. A substance can bioaccumulate in a food chain to toxic concentration, and some can cause eutrophication (IaS4). A change in the size of a population will affect other populations in the same community. The distribution and abundance of organisms, and changing conditions, within an ecosystem can be investigated using techniques including: identification keys; transects and quadrats; capture, mark, release and recapture; sampling indicator species; and using instruments to measure abiotic factors such as temperature, light intensity, soil moisture and pH.	1. explain how some abiotic and biotic factors affect communities, including environmental conditions, toxic chemicals, availability of food and other resources, and the presence of predators and pathogens
	2. describe how to carry out a field investigation into the distribution and abundance of organisms in an ecosystem and explain how to determine their numbers in a given area M2d PAG3
	3. in the context of data related to organisms within a population: a) calculate arithmetic means M2b, M2f b) use fractions and percentages M1c c) plot and draw appropriate graphs selecting appropriate scales for the axes M4a, M4c d) extract and interpret information from charts, graphs and tables M2c

Linked learning opportunities

Practical work:

- investigate the distribution and abundance of organisms in an ecosystem

Ideas about Science:

- bioaccumulation and eutrophication as unintended impacts of human activity on the environment (IaS4)